

§ 3.12.

P1. Solution: $A = \begin{bmatrix} 4 & 5 \\ -2 & -2 \end{bmatrix}$, $f(t) = \begin{bmatrix} 4e^t \cos t \\ 0 \end{bmatrix}$

Using the eigenvalue - eigenvector method, you can find

$$e^{At} = e^t \begin{bmatrix} \cos t + 3 \sin t & 5 \sin t \\ -2 \sin t & \cos t - 3 \sin t \end{bmatrix}$$

Then, using $x(0) = x^0 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$, we get

$$\begin{aligned} x(t) &= e^{At} x^0 + \int_0^t e^{A(t-s)} f(s) ds \\ &= e^{At} \int_0^t e^{-s} \begin{bmatrix} \cos s - 3 \sin s & -5 \sin s \\ 2 \sin s & \cos s + 3 \sin s \end{bmatrix} \begin{bmatrix} 4e^s \cos s \\ 0 \end{bmatrix} ds \\ &= e^{At} \int_0^t \begin{bmatrix} 4 \cos^2 s - 12 \sin s \cos s \\ 8 \sin s \cos s \end{bmatrix} ds \\ &= e^{At} \begin{bmatrix} \sin 2t + 3 \cos 2t + 2t - 3 \\ -2 \cos 2t + 2 \end{bmatrix} \\ &= e^{At} \begin{bmatrix} 2 \sin t \cos t - 6 \sin^2 t + 2t \\ 4 \sin^2 t \end{bmatrix} \\ &= e^t \begin{bmatrix} 2t \cos t + 6t \sin t + 2 \sin t \\ -2t \sin t \end{bmatrix} \end{aligned}$$

P2. Solution: $A = \begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix}$, $f(t) = \begin{bmatrix} e^t \\ e^t \end{bmatrix}$, $x^0 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$$e^{At} = e^t \begin{bmatrix} 1+2t & -4t \\ t & 1-2t \end{bmatrix}$$

then

$$\begin{aligned} x(t) &= e^{At} x^0 + \int_0^t e^{A(t-s)} f(s) ds \\ &= e^{At} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + e^{At} \int_0^t e^{-s} \begin{bmatrix} 1-2s & 4s \\ -s & 1+2s \end{bmatrix} \begin{bmatrix} e^s \\ e^s \end{bmatrix} ds \\ &= e^{At} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + e^{At} \begin{bmatrix} t+t^2 \\ t+\frac{t^2}{2} \end{bmatrix} \\ &= e^t \begin{bmatrix} 1-2t \\ 1-t \end{bmatrix} + e^t \begin{bmatrix} t-t^2 \\ t-\frac{t^2}{2} \end{bmatrix} = e^t \begin{bmatrix} 1-t-t^2 \\ 1-\frac{t^2}{2} \end{bmatrix} \end{aligned}$$

P3. Solution: $A = \begin{bmatrix} 2 & -5 \\ 1 & -2 \end{bmatrix}$, $f(t) = \begin{pmatrix} \sin t \\ \tan t \end{pmatrix}$

$$e^{At} = \begin{bmatrix} \cos t + 2 \sin t & -5 \sin t \\ \sin t & \cos t - 2 \sin t \end{bmatrix}$$

$x(0) = x^0 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, Then

we get

$$\begin{aligned} x(t) &= e^{At} x^0 + \int_0^t e^{A(t-s)} f(s) ds \\ &= e^{At} \int_0^t \begin{bmatrix} \cos s - 2 \sin s & 5 \sin s \\ -\sin s & \cos s + 2 \sin s \end{bmatrix} \begin{bmatrix} \sin s \\ \tan s \end{bmatrix} ds \\ &= e^{At} \int_0^t \begin{bmatrix} \cos s \sin s - 2 \sin^2 s + 5 \sin s \tan s \\ -\sin^2 s + \sin s + 2 \sin s \tan s \end{bmatrix} ds \\ &= \text{[see P.569]} \end{aligned}$$